OC Core 1.3 Journal Core: A Source-Audited Revalidation of Collapse, Residue, and Rebirth

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Abstract

This article revalidates a bounded kernel of the Ontology of Continua directly from the tagged Core 1.2 LaTeX sources and the locked December 11, 2025 PDF under a strict source-audit discipline. The positive claim is narrower but stronger: the surviving kernel supports a source-audited classification theorem on post-collapse identity. When a continuum loses every admissible realization, the original continuum is dead, structural residue may remain, and any later live continuation grounded in that residue must be treated as rebirth rather than persistence of the original continuum. Figure 1 renders the grammar; Figure 2 shows the support boundary that keeps the claim externally defensible. The contribution is both scientific and methodological. Scientifically, the paper restores a lawful theorem chain from the public Core 1.2 source corpus instead of relying on optimistic summary layers. Methodologically, it shows how unresolved theorem families are excluded rather than rhetorically blended into a stronger claim. The result is a bounded foundations article with a fully exposed support boundary.

Problem Statement

Can the surviving source-native OC Core kernel support a bounded article on collapse, residue, and rebirth under a strict source-audit discipline? The problem is not whether the entire inherited OC corpus can be restated attractively. The real question is whether a bounded subset of source-native claims survives a hostile audit strongly enough to support a standalone mathematical foundations paper. That distinction matters because the inherited package mixed preserved axioms, preserved definitions, reformulated theorem families, and removed noncanonical artefacts into one outward-facing narrative. The manuscript validates only the source-native rows that survive the repaired canon and dependency audit and feed the main proof. The manuscript does not claim a generally defended axis/state lift theorem or full external readiness of the inherited OC Core theorem corpus. Accordingly, the problem is solved only if the main theorem chain can be reconstructed from source-native material that remains canon-clean after repair, if the argument exposes where the defended case ends, and if the paper remains falsifiable in the precise sense that renewed audit or contrary formal analysis could overturn it.

Main Result

The main result is a bounded source-audited classification theorem about lawful post-collapse identity, not a universal completion of the inherited ontology.

Theorem A (source-audited derived post-collapse identity classification). Let K be a continuum in the audited OC Core kernel. If K loses every admissible realization under its current threshold and embedding regime, then the original continuum is dead, any surviving post-collapse structure is to be classified as residue rather than as a still-live instance of K, and any later live continuum grounded in that residue is to be classified as rebirth rather than persistence of K.

The manuscript carries this result in revised form: Within the surviving source-native OC Core kernel, once a continuum loses every admissible realization, the original continuum is dead, any surviving post-collapse structure is residue rather than a still-live instance of that continuum, and any later live continuum grounded in that residue is rebirth rather than persistence of the original one. The paper does not present Theorem A as a new primitive source theorem. It presents it as a derived classification result whose substantive force comes from the restored death theorem and the restored irreversibility corollary; Definitions 12.5 and 12.6 do not function as independent theorem sources, but they do fix the classification criteria and lawful terminology for the two non-persistence cases. The main theorem chain consists of 3 revised main results, 3 derived lemmas, 2 supporting axioms, and 5 supporting definitions reproduced in the article. The inherited theorem families that formerly underwrote broader lift or representability claims are not used here unless they survive separate revalidation. The paper therefore asks the reader to evaluate one cleanly bounded source-native argument. It does not ask the reader to accept the whole inherited framework on trust, and it does not smuggle open theorem families through stylistic compression.

Definitions and Quantitative Observables

The manuscript fixes four load-bearing notions: admissible realization, death, residue, and rebirth. Admissible realization is the nonempty state region that keeps a continuum lawfully live under its current thresholds and embedding regime. Death is the loss of that admissible realization. Residue is the structural trace that can remain after death without preserving live identity. Rebirth is the emergence of a later live continuum grounded in that trace but not identical with the original continuum. The article also keeps a small operational vocabulary for later projections. These observables are not used to inflate the present argument into a full universal law; they are included because they specify what later mathematical or domain-level projections would have to measure if the bounded theorem grammar is taken seriously. These observables matter because they turn the argument from an ontological slogan into a falsifiable grammar. They tell the reader what kind of counterexample would matter, what kind of terminal loss counts as death, and what kind of post-collapse persistence claim is prohibited by the manuscript. - `contradiction_load`: Minimal mismatch between required differences and currently expressible axes / admissible coordinates. - `minimal_lift_rank`: Smallest number of new axes or auxiliary state coordinates required to restore expressivity without changing the declared contradiction packet. - `collapse_threshold`: Boundary at which no admissible realization survives under the current thresholds and embedding constraints. - `admissible_volume_or_fiber_survival_measure`: Positive-size witness of surviving admissible states or surviving exact fibers; zero at collapse. - `stabilization_cost`: Margin between current contradiction load and the nearest collapse or lift-trigger threshold.

Source Audit and Canon Repair

The manuscript is source-native by construction. The main theorem chain is rebuilt from the tagged Core 1.2 LaTeX repository and the canonical compiled PDF, not from later optimistic summaries. Every load-bearing row in the main argument is required to survive three linked repairs before it can carry weight: source witnessing, canonical uniqueness, and dependency exactness. This is the central methodological discipline of the paper. The tagged public source snapshot is v1.2.1 at 2c61b7879ed36cd8366d87892a066082b6418ce8. For the bounded surviving kernel, the theorem-bearing draft sources are concentrated in the axiom, core-results, and collapse/rebirth source modules of that public release. The PDF serves as the compiled witness for page-line anchors on the positive source rows, while the revised main theorem is tracked explicitly as a bounded classification theorem whose support is distributed across those anchored rows rather than disguised as a verbatim single-source theorem. Companion dossiers extend the article with the full source witness matrix, theorem-support ledger, chronology dossier, and projection atlases, but none of those companions relax the rule that the main theorem chain must be carried by the audited Core 1.2 kernel alone. That repair changes the scientific meaning of publication readiness. A row cannot remain theorem-admissible merely because it once appeared inside an optimistic summary layer. If a row loses unique canonical binding, depends on unresolved rebinding, or is removed from the surviving OC theory, the row is excluded from the main article argument. In the present manuscript, that discipline removes the inherited lift-heavy theorem family from the defended case and leaves a narrower but cleaner theorem grammar. The repaired corpus is therefore best read as an explicit partition of admissible source material. Some rows survive and support the main theorem chain. Some rows remain scientifically interesting but require separate revalidation. Others are removed from the present article. This manuscript treats that partition as part of the scientific contribution rather than as editorial housekeeping. Representative source-native witnesses used directly in the present bounded argument are the following: - Theorem A (source-audited post-collapse identity classification): bounded classification theorem proved from the audited source-native chain Source Theorem 3 (Death through boundary and embedding collapse), Source Corollary 3.2 (Irreversibility of death), Axiom 3.2 (Death condition), Axiom 3.3 (Irreversibility of death), Definition 12.1 (Collapse), Definition 12.3 (Internal collapse), Definition 12.4 (External collapse), Definition 12.5 (Residue), Definition 12.6 (Rebirth). - Lemma 1 (Collapse entails death): derived from the audited source-native support chain Source Theorem 3 (Death through boundary and embedding collapse), Axiom 3.2 (Death condition), Definition 12.1 (Collapse), Definition 12.3 (Internal collapse), Definition 12.4 (External collapse). - Lemma 2 (Post-collapse trace is residue): derived from the audited source-native support chain Source Theorem 3 (Death through boundary and embedding collapse), Axiom 3.2 (Death condition), Definition 12.1 (Collapse), Definition 12.3 (Internal collapse), Definition 12.4 (External collapse), Definition 12.5 (Residue). - Lemma 3 (Later live continuum is rebirth): derived from the audited source-native support chain Source Corollary 3.2 (Irreversibility of death), Axiom 3.3 (Irreversibility of death), Definition 12.5 (Residue), Definition 12.6 (Rebirth). - Definition 12.1 (Collapse): A continuum K collapses at time t* if the admissible state space becomes empty, continuumness decays to zero, and every candidate configuration violates at least one existence or stability threshold. -

Definition 12.3 (Internal collapse): Internal collapse occurs when collapse is caused solely by the internal evolution operator E = (F, G, H, Q, R, S, U) acting on K while the embedding space M remains structurally static over the relevant interval. - Definition 12.4 (External collapse): External collapse occurs when the embedding space M changes so that no configuration of K remains compatible with the new constraints, even though the prior internal evolution stayed within the previously admissible region. - Definition 12.5 (Residue): The residue of a collapsed continuum K is the set of structures in the embedding space that remain after Omega(K) = empty; residues may persist without preserving the original live identity. - Definition 12.6 (Rebirth): A rebirth event occurs when, after the collapse of K, a new continuum K' appears while Omega(K) = empty and k(K,t) = 0, inherits at least one structural element from residue, and satisfies the birth conditions for some level. - Axiom 3.2 (Death condition): A continuum dies at time t* when Omega(K(t*)) = empty; after death the operators F, G, H, Q, R, S, and U are no longer defined for that continuum. - Axiom 3.3 (Irreversibility of death): No operator acting within the same level can restore a dead continuum; any new live continuum is considered a new entity. - Source Theorem 3 (Death through boundary and embedding collapse): derived from the audited source-native support chain Axiom 3.2 (Death condition), Definition 12.1 (Collapse), Definition 12.3 (Internal collapse), Definition 12.4 (External collapse). - Source Corollary 3.2 (Irreversibility of death): derived from the audited source-native support chain Axiom 3.3 (Irreversibility of death), Definition 12.6 (Rebirth).

Theorem Ladder and Admissible Support

The positive proof architecture is organized as a ladder instead of a slogan because the manuscript wants the reader to see exactly where the result becomes legal. Stage 1 fixes death as loss of admissible realization. Stage 2 adds irreversibility, ruling out same-level restoration of the dead continuum as the same entity. Stage 3 adds residue as a trace-preserving but identity-nonpreserving aftermath. Stage 4 adds rebirth as the lawful semantics of any later continuation. Stage 5 composes those source-native rows into one audited classification theorem. This ladder is important for two reasons. First, it prevents the paper from hiding a weak middle step. Second, it makes the argument readable to editors and reviewers who need to know whether the paper is really proving one proposition or quietly shifting across several incompatible meanings of collapse. - Once admissible realization is empty, the original continuum is dead and no longer lawfully live. - Irreversibility forbids same-level restoration of the dead continuum as the same entity. - Residue records which structural traces survive after collapse without preserving the original continuum as live. - Any later continuation after collapse is rebirth of a new continuum from residue, not persistence of the original one. - Together, the restored death theorem, irreversibility corollary, and residue/rebirth definitions yield a bounded classification theorem: after admissible-state loss, persistence ends, residue may remain, and any later live continuation is rebirth rather than survival of the original continuum.

The supporting source rows used in the article are the following: - Theorem A (source-audited post-collapse identity classification): Bounded theorem stating that death, residue, and rebirth semantics survive source-native revalidation. - Definition 12.1 (Collapse): Collapse is defined by the loss of a nonempty admissible state space rather than by metaphorical failure language. - Definition 12.3 (Internal collapse): Internal collapse isolates endogenously generated destructive evolution as one lawful terminal mechanism. - Definition 12.4 (External collapse): External collapse isolates embedding-driven incompatibility as a distinct lawful terminal mechanism. - Definition 12.5 (Residue): Definition 12.5 names the surviving post-collapse trace as residue once the death theorem has already ruled out persistence of the original live continuum. - Definition 12.6 (Rebirth): Definition 12.6 names any later live continuation grounded in residue as rebirth once the irreversibility corollary has already ruled out restoration of the original entity. - Axiom 3.2 (Death condition): Death is fixed by the source-native death condition once admissible realization becomes empty. - Axiom 3.3 (Irreversibility of death): Irreversibility forbids restoration of the dead continuum as the same entity. - Source Theorem 3 (Death through boundary and embedding collapse): Exact source-native theorem witness for death through admissible-state collapse in the restored Core 1.2 results corpus. - Source Corollary 3.2 (Irreversibility of death): Exact source-native corollary witness that death is irreversible and any apparent recovery is the birth of a new continuum.

The derived proof-spine rows promoted into the manuscript are the following: - Lemma 1 (Collapse entails death): If the admissible state space becomes empty, the original continuum is dead and cannot lawfully persist as a live continuum. - Lemma 2 (Post-collapse trace is residue): After collapse, any surviving structure is residue rather than a still-live instance of the original continuum. - Lemma 3 (Later live continuum is rebirth): Any later live continuum grounded in residue is rebirth of a new continuum rather than persistence of the original one.

Lemma Chain and Proof of Theorem A

The manuscript does not treat Theorem A as an unsupported rhetorical synthesis. It is proved as a bounded source-audited classification theorem: a short lemma chain extracted from the surviving kernel is composed into one explicit main result whose source provenance is fully exposed in the companion materials. - If the admissible state space becomes empty, the original continuum is dead and cannot lawfully persist as a live continuum. - After collapse, any surviving structure is residue rather than a still-live instance of the original continuum. - Any later live continuum grounded in residue is rebirth of a new continuum rather than persistence of the original one. Proof of Lemma 1. Axiom 3.2 gives the death condition directly, and Source Theorem 3 in the restored Core 1.2 results corpus promotes that axiom into an explicit theorem: a continuum dies when its admissible realization becomes empty. Definition 12.1 strengthens that terminal event by adding vanishing continuumness and universal threshold failure, while Definitions 12.3 and 12.4 show that the same terminal semantics holds for both internal and external routes to collapse. Therefore the empty admissible set is not a temporary excursion; it is lawful terminal loss of the original live continuum. Proof of Lemma 2. Lemma 1 has already established that collapse ends the life of the original continuum. Source Theorem 3 therefore rules out any reading on which a surviving post-collapse structure is still the original live object. Definitions 12.1 and 12.5 then fix the lawful classification criterion for the remaining trace: it may persist structurally in the embedding space, but only as residue and not as continued live identity. Proof of Lemma 3. Source Corollary 3.2 together with Axiom 3.3 excludes any same-level or identity-preserving restoration of a dead continuum. Once that prohibition is in place, any later live continuum grounded in post-collapse trace must be numerically new rather than persistent continuation of the original one. Definition 12.6 fixes the lawful classification of exactly that later live case as rebirth. Proof of Theorem A. Lemma 1 establishes the death of the original continuum after admissible-space collapse. Lemma 2 then classifies any surviving structure as post-collapse residue rather than continued live identity. Lemma 3 shows that any later live continuation grounded in that trace must be numerically new and must therefore be classified as rebirth. Theorem A is thus a derived classification theorem reconstructed from one restored theorem, one restored corollary, and the source-native classification criteria for residue and rebirth; it is not offered as a free-standing primitive theorem beyond the surviving source corpus. No excluded lift theorem is used, no hidden representability family is imported, and no stronger transition law is presumed beyond the bounded post-collapse identity grammar stated above.

Proof Architecture

The proof is conservative in the strongest useful sense. It does not attempt to reconstruct the whole inherited root law from whatever remains salvageable. Instead it proves that a smaller, cleaner post-collapse identity grammar already follows from the surviving kernel. The argument begins with the restored death theorem and irreversibility corollary, because those are the exact source-native rows that make terminal loss lawful. Residue and rebirth are then added as semantic controls on what can and cannot be said afterward. A second layer of the proof is exclusionary rather than additive. The manuscript explicitly refuses to use unrevalidated lift-support theorems as hidden bridges from contradiction to dimensional growth. This is not a loss of rigor but an increase in rigor. By forcing the positive paper to live without those supports, the argument reveals exactly which part of the root law is externally defensible now and which part still belongs to future work. A third layer is editorially crucial: the manuscript keeps preserved source support, revised theorem wording, and excluded inherited theorem families visibly distinct. The reader can therefore audit the distance between surviving source support, revised wording, and excluded theorem families without having to reconstruct the whole developmental history of the project.

Appendix-Only and Excluded Support Boundary

The present route does not keep a shadow theorem body in reserve. Any inherited theorem family that would be needed to defend a general contradiction-to-lift claim is outside the main article unless it survives separate revalidation with explicit provenance and exact dependency support. At the current manuscript boundary, no theorem family remains in a provisional support limbo. The inherited lift and representability families are excluded from the main article rather than presented as tacitly available support.

Literature Screen

The paper is not written as if it were the first text ever to discuss collapse, viability, or regime change. Its literature screen identifies the nearest formal baselines, states what each contributes, and explains exactly why none of them substitutes for the source-native OC theorem chain. - YASHIN_OC_CORE_2025: Primary archival witness for the exact PDF theorem corpus audited by the present article. Alexander Yashin, Ontology of Continua - Core v1.2.0, Zenodo, 2025, DOI 10.5281/zenodo.17903912, https://zenodo.org/records/17903912. - OC_CORE_PUBLIC_TAG_V121: Public LaTeX source lock used to witness exact theorem, axiom, and definition bodies in the audited OC corpus. Alexander Yashin, ontology-of-continua-core-main, GitHub repository, tag v1.2.1, commit 2c61b7879ed36cd8366d87892a066082b6418ce8, https://github.com/alexanderyashin/ontology-of-continua-core-main. - GUARINO_1998: External baseline for formal ontological category discipline, but not a theorem source for the death-residue-rebirth identity law defended here. Nicola Guarino, Formal Ontology and Information Systems, in Formal Ontology in Information Systems, IOS Press, 1998. - GRENON_SMITH_2004: Nearest cited formal-ontology treatment of dynamic identity and process structure, but not a source-audited theorem chain for admissible-state collapse. Pierre Grenon and Barry Smith, SNAP and SPAN: Towards Dynamic Spatial Ontology, Spatial Cognition and Computation 4(1), 2004. - LOWE_2006: External baseline for metaphysical identity and persistence categories, but not for collapse-triggered residue and rebirth semantics. E. J. Lowe, The Four-Category Ontology: A Metaphysical Foundation for Natural Science, Oxford University Press, 2006. - THOM_1994: External baseline for structural transition language; it studies regime change classes but does not supply the identity grammar of death, residue, and rebirth proved here. Rene Thom, Structural Stability and Morphogenesis: An Outline of a General Theory of Models, Westview Press, 1994. - AUBIN_1991: Nearest external baseline for admissible-state survival under constraints, but it does not define residue and rebirth as post-collapse identity classes. Jean-Pierre Aubin, Viability Theory, Birkhauser, 1991. - STROGATZ_2015: External baseline for bifurcation and qualitative regime change, but not a proof source for lawful post-collapse identity semantics. Steven H. Strogatz, Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering, 2nd ed., Westview Press, 2015. - CONLEY_1978: External baseline for terminal invariant-set structure and loss of persistence, but not for the unified death-residue-rebirth law grammar of the OC kernel. Charles Conley, Isolated Invariant Sets and the Morse Index, American Mathematical Society, 1978. - MILNOR_1985: External baseline for persistence and new post-transition objects, but it does not package residue and rebirth into a single source-audited theorem chain. John Milnor, On the concept of attractor, Communications in Mathematical Physics 99 (1985), 177-195. Among those comparators, viability theory is the nearest external baseline for nonempty admissible-state survival under constraints; Conley and Milnor are the nearest baselines for persistence failure and the appearance of new post-transition objects; Thom and Strogatz provide the nearest language for qualitative transition classes. None of those baselines, however, proves the exact identity trichotomy defended here: death after admissible-state loss, residue as surviving trace without live identity, and rebirth as a new continuum rather than resurrection. | comparator | nearest formal role | agreement | bounded difference | missing component | | — | — | — | — | — | | GUARINO_1998_AND_GRENON_SMITH_2004 | formal ontology of identity and change | Formal ontology provides the closest cited baseline for disciplined category vocabulary, identity conditions, and the distinction between stable objects and changing processes. | These frameworks do not prove the specific post-collapse identity trichotomy defended here and do not derive death, residue, and rebirth from admissible-state loss under a source-audited theorem chain. | no theorem-level rule that admissible-state loss kills the original object, leaves only residue, and makes any later live continuation a numerically new entity | | LOWE_2006 | metaphysical identity and persistence categories | Metaphysical category theory supplies a cited baseline for identity discipline and for distinguishing what a thing is from how it persists through change. | It does not furnish the explicit collapse-triggered law grammar used here, and it does not connect persistence failure to residue and rebirth via theorem-level source witnesses. | no source-audited theorem that turns persistence failure after admissible-state loss into a death-residue-rebirth classification | | AUBIN_1991 | admissible-state survival under constraints | Nonempty admissible sets and threshold-governed viability determine whether a structured object remains lawfully live. | Viability theory does not distinguish residue and rebirth as post-collapse identity classes for the same object. | no source-audited rule that any later live structure grounded in post-collapse trace is a new entity rather than persistence | | CONLEY_1978_AND_MILNOR_1985 | persistence failure and new post-transition objects | Invariant-set and attractor theory supply the nearest external language for terminal loss of persistence and the appearance of new dynamical objects after transition. | They do not package death, residue, and rebirth into one source-audited identity theorem with a source-native witness chain. | no bounded theorem separating dead object, surviving trace, and later new object inside one audited identity grammar | | THOM_1994_AND_STROGATZ_2015 | qualitative transition taxonomy | These frameworks provide the closest broad language for structural transition, bifurcation, and regime change. | They do not supply the theorem-level identity law used here to separate collapse, surviving residue, and lawful rebirth of a new continuum. | no post-collapse identity classification theorem linked to explicit source-native death and irreversibility witnesses | The source witness matrix, theorem-support dossier, and supplementary appendix accompany the article as supporting materials rather than as bibliography items. This literature screen also bounds the novelty claim. The paper does not claim to supersede catastrophe theory, viability theory, invariant-set theory, or nonlinear dynamics. Its comparative claim is narrower and sharper. Catastrophe theory analyzes transition classes, viability theory studies survival under constraints, and invariant-set or attractor theory studies persistence failure and post-transition objects. The present article instead makes explicit, within the OC family, a source-audited derived classification that separates death, residue, and rebirth as distinct identity statuses under one strictly audited chain of support.

Novelty and Prior-Art Boundary

Within the surviving source-native OC Core kernel, once a continuum loses every admissible realization, the original continuum is dead, any surviving post-collapse structure is residue rather than a still-live instance of that continuum, and any later live continuum grounded in that residue is rebirth rather than persistence of the original one. The main proof does not claim a universal contradiction-to-lift theorem, a full representability theory, or completed universal closure. It claims only a source-audited classification theorem on death, residue, and rebirth after admissible-state loss. The novelty claim is therefore bounded in theorem scope and deliberately modest. Relative to the inherited OC corpus, the manuscript contributes a conservative source-audit repair that separates surviving theorem support from removed or unrevalidated material and shows that the surviving source theorem plus irreversibility corollary already force a post-collapse non-persistence classification. Relative to broader foundational discourse, it does not claim priority for every idea involving collapse, persistence failure, or structural rebirth. Its positive contribution is narrower and cleaner: a source-audited derived classification after admissible-state loss, together with a reproducible method for rebuilding such a classification from heterogeneous theoretical corpora without silently importing unresolved premises. The closest theorem-level comparator is the viability-kernel baseline: it supplies the nearest external criterion for whether admissible evolution still exists under constraints, but it stops at survival versus nonsurvival. The article-level contribution begins exactly where that baseline stops by making the post-emptiness identity consequences explicit and auditable. | external baseline | closest formal conclusion | theorem-level gap left open externally | added article-level content in Theorem A | | — | — | — | — | | Aubin viability-kernel result class | nonempty admissible evolution distinguishes viable from nonviable states under constraints | no theorem that emptiness kills the original object, leaves only residue, and forces any later live continuation to count as a new object | the article makes those post-emptiness identity consequences explicit by combining the restored death theorem, the restored irreversibility corollary, and the naming definitions for residue and rebirth | | Conley-Milnor persistence-failure language | transition may destroy persistence and produce later dynamical objects | no single audited identity theorem connecting death, surviving trace, and later new object under one source-native support chain | the article packages those three statuses into one bounded derived classification result with explicit source witnesses and an exposed exclusion boundary | - Closest external trigger-level baseline: Aubin's viability-kernel result class distinguishes states from which at least one admissible trajectory remains inside the constraint set from states outside that kernel. - Theorem-level gap: once the admissible set is empty, that baseline no longer classifies the identity status of the original object, the status of any surviving structural trace, or the status of any later admissible continuation. - Added article-level explicitness in Theorem A: the surviving source theorem, the irreversibility corollary, and the residue/rebirth naming definitions are assembled into one auditable post-collapse classification, so admissible-state emptiness is read as death of the original continuum, surviving structure is read as residue rather than survival, and any later live continuation grounded in that trace is read as rebirth of a numerically new continuum. - Secondary comparison lane: Conley-Milnor language tracks persistence failure and new post-transition objects, but it does not package death, residue, and rebirth into one audited identity theorem tied to explicit source-native witnesses. That boundary is essential for honest journal positioning. The paper is not strong because it says everything. It is strong because it says exactly what is presently defensible, states why stronger inherited claims are withheld, and converts that restraint into a reproducible methodological contribution that a hostile reader can audit line by line.

Cross-Domain Interpretation and Scope Discipline

The manuscript has implications beyond a single formal kernel, but it does not claim that those implications are already proved as a universal cross-domain theorem. Table 1 therefore functions as a bounded projection map rather than as proof of complete downstream closure. Mathematics, domain science, and later domain families are shown as projection targets whose admissibility depends on separate theorem programs and empirical or formal validation routes. This distinction is important for editorial honesty. A generalist venue needs to understand why the work matters beyond one notation system, but it also needs to see that the manuscript does not inflate projection potential into accomplished proof. The paper therefore uses cross-domain language only to identify where the collapse-residue-rebirth grammar should be tested next and what would count as a failed projection. Tier 1 mathematics remains outside the main theorem chain in the present article, and the cross-domain atlas is used only to mark later test lanes and downstream research families without silently promoting them into proved support. In that sense the paper offers a general law grammar and a disciplined research agenda at the same time. It does not confuse the one with the other.

Reviewer Objections and Resolutions

The manuscript includes a concrete objection-resolution layer because recurring scientific objections should be surfaced before they return as review failures. - Objection: The inherited root slogan overreaches by treating contradiction-driven lift and collapse as equally defended in the current corpus. Resolution: The article restricts the defended claim to collapse, residue, and rebirth and explicitly excludes the general lift branch from the theorem it asks the reader to accept. - Objection: Optimistic summary layers can mix surviving, removed, and unresolved rows into one apparent theorem body. Resolution: Only canon-clean and dependency-exact rows survive into the manuscript, and excluded rows remain outside the main theorem chain. - Objection: Collapse language can become metaphorical unless it is tied to admissible-state loss and identity rules. Resolution: The manuscript binds collapse to admissible-space emptiness, residue to post-collapse structural trace, and rebirth to post-collapse continuation without identity preservation. - Objection: A foundations paper can overclaim cross-domain closure by narrating future projections as accomplished proof. Resolution: Cross-domain material is carried only as a bounded projection map and explicit future test agenda, never as completed proof.

Falsifiability and Failure Modes

The article should be rejected if any of the following conditions is met: - a positive-body source row loses canonical uniqueness or dependency exactness under renewed audit; - an admissible-space collapse is shown to occur without the residue and rebirth distinctions used in the positive theorem grammar; - a later continuation can be identified with the original continuum without violating the manuscript's own collapse semantics; - an excluded or unrevalidated theorem family is required in order for Theorem A to go through; - the projection table is found to overstate downstream closure rather than marking bounded future tests.

Limitations and Nonclaims

- The manuscript does not claim universal closure of the full inherited OC Core 1.2 theorem corpus.
- It does not claim that a general contradiction-driven axis or state lift theorem is already externally defended in the main proof.
- It does not claim that the projection families listed in Table 1 are already proved across mathematics, natural science, and broader domain families.
- It does not ask the reader to equate exclusion from the main proof with scientific uselessness; several excluded theorem families remain valid revalidation targets for later papers.

Evidence and Reproducibility

The present article is anchored to the source-native PDF ‘Ontology of Continua — Core 1.2’ (Version 1.2 — December 11, 2025) and to the tagged public repository snapshot v1.2.1. The evidentiary center of gravity is not prose elegance but source fidelity under repaired canon constraints. A complete reproduction therefore requires four checks. First, the source-native claim inventory must reproduce the same surviving rows. Second, the canon repair must reproduce the same unique bindings and exact dependency outcomes for those rows. Third, the positive theorem ladder must still be reconstructible without calling excluded theorem families back into service. Fourth, the figure and table assets must regenerate from the same canonical inputs with the same captions, provenance, and bounded interpretation. Because the manuscript is intentionally conservative, reproducibility includes the possibility of reproducing failure. If a renewed audit changes the status of a support row, the manuscript is supposed to change with it. That is a scientific feature of the current method, not an embarrassment to be hidden. The same source-native basis also preserves the explicit declared limitations already present in Core 1.2. These limitations are not editorial noise; they are part of the truth conditions under which the present bounded theorem is submitted: - Bridging from the structural language of OC to concrete datasets remains nontrivial and requires domain-specific modelling. - Quantitative models for Kx -> Kx+1 transitions remain incomplete and are only available in partial case-specific form. - Threshold taxonomy is structurally complete, but explicit functional forms still have to be specified separately for each system class. - Operational measures of expressive capacity for realistic cognitive, social and theoretical systems are not yet fully developed. - Inter-continuum interactions are structurally represented, but quantitative theories for complex coupled continua remain largely schematic.

Figures and Table Callouts

- Figure 1 summarizes the bounded law of collapse, residue, and rebirth, and marks the lift branch as outside the present main theorem chain.
- Figure 2 shows the theorem ladder together with the support-status discipline that separates surviving support from excluded inherited theorem families.
- Table 1 maps the article to later publication families and domain projections without overstating present closure.

References

[1] Alexander Yashin, Ontology of Continua - Core v1.2.0, Zenodo, 2025, DOI 10.5281/zenodo.17903912, https://zenodo.org/records/17903912. [2] Alexander Yashin, ontology-of-continua-core-main, GitHub repository, tag v1.2.1, commit 2c61b7879ed36cd8366d87892a066082b6418ce8, https://github.com/alexanderyashin/ontology-of-continua-core-main. [3] Nicola Guarino, Formal Ontology and Information Systems, in Formal Ontology in Information Systems, IOS Press, 1998. [4] Pierre Grenon and Barry Smith, SNAP and SPAN: Towards Dynamic Spatial Ontology, Spatial Cognition and Computation 4(1), 2004. [5] E. J. Lowe, The Four-Category Ontology: A Metaphysical Foundation for Natural Science, Oxford University Press, 2006. [6] Rene Thom, Structural Stability and Morphogenesis: An Outline of a General Theory of Models, Westview Press, 1994. [7] Jean-Pierre Aubin, Viability Theory, Birkhauser, 1991. [8] Steven H. Strogatz, Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering, 2nd ed., Westview Press, 2015. [9] Charles Conley, Isolated Invariant Sets and the Morse Index, American Mathematical Society, 1978. [10] John Milnor, On the concept of attractor, Communications in Mathematical Physics 99 (1985), 177-195.